THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–8) | Class 7 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** 6,04,800	1 day = 86,400 seconds.
2	**(a)** 15	Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction.
3	**(b)** 0.024	Multiply two at a time and count the decimal places carefully.
4	**(b)** 21	Square first, then multiply.
5	**(d)** 90°	Reason with the angle relations.
6	**(d)** 108	Find the first multiple of 9.
7	**(a)** 3	List the triples $a \le b \le c$ with $a + b > c$.
8	**(b)** $\frac{1}{4}$	Work the bracket first.
9	**(c)** 7	Count zeros in the product.
10	**(d)** 15	Sum first, then double, then subtract.
11	**(c)** 48	Divide the larger number by the smaller.
12	**(a)** $2x + 13$	Expand each bracket, watching the signs.
13	**(d)** 120°	Share 180° in the given ratio.
14	**(d)** 9873	Start from 9876 and test downwards.
15	**(b)** 40°	Base angle $= 180^\circ - e$; apex $= 180^\circ - 2\times$ base.
16	**(b)** 135	Find the number first.
17	**(a)** Both A and R are true and R is the correct explanation of A.	1,00,00,000 / 1,00,000 = 100.
18	**(a)** Both A and R are true and R is the correct explanation of A.	3 x 4 = 12 and 5 + 12 = 17.
19	**(a)** Both A and R are true and R is the correct explanation of A.	0.50 > 0.45.
20	**(a)** Both A and R are true and R is the correct explanation of A.	Three times n is 3n, and five more is 3n + 5.

SECTION — B (Very Short Answer) — Answers 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	1 day = 86,400 seconds (1); $7 \times 86400 = 6,04,800$; international: 604,800 (1).	1+1
22	Innermost brackets; Next level (1); Final operations; Value $= 15$ (1).	1+1
23	The missing number is the product divided by the known factor (1); $\square = 0.162 \div 0.6 = 0.27$ (1).	1+1
24	Expand: $7x - 14 - 3x + 15$ (1); Answer: $4x + 1$ (1).	1+1
25	Let the angle be y: $180 - y = 3(90 - y)$; $2y = 90$ (1); $y = 45^\circ$ (1).	1+1
26	$11(a + b) = 110$ so $a + b = 10$; $9(a - b) = 36$ so $a - b = 4$ (1); $a = 7$, $b = 3$; Number $= 73$ (1).	1+1

SECTION — C (Short Answer) — Answers 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Total $= 2,50,000 \times 48 = 1,20,00,000$ (1); Words: one crore twenty lakh (1); Nearest crore: $1,00,00,000$ (1).	1+1+1
28	Full sets $= 24 \div 6 = 4$; Discount $= 4 \times 20 = 80$ (1); Total $= 24 \times 35 + 30 \times 12 - 80 = 1120$ (1); Check with units (₹) (1).	1+1+1
29	$2.5 \times 0.4 = 1$ (1); $1.6 \div 0.4 = 4$ (1); Sum $= 5$ (1).	1+1+1
30	$P = 2[(5x + 4) + (3x - 1)]$; $= 2(8x + 3)$ (1); $= 16x + 6$ (1); At $x = 6$: 102 cm (1).	1+1+1
31	Angles on a straight line add up to 180° (1); Fourth angle $= 180^\circ - (35 + 45 + 50)^\circ = 50^\circ$ (1); Kinds: 35° acute, 45° acute, 50° acute, 50° acute (1).	1+1+1
	OR: Digit sum $= 13 + 2 \times *$; Need a multiple of 9 (1); Possible $*$: 7 (1); Numbers: 47376 (1).	1+1+1

32	Exterior angles at B and C add up to $180^\circ + \angle A$ because $(180 - B) + (180 - C) = 360 - (180 - A)$; $4x + 60 + 3x + 100 = 230$ (1); $x = 10$; Exterior angles 100° and 130° (1); $\angle B = 80^\circ$, $\angle C = 50^\circ$ (1).	1+1+1
	OR: Ravi's one day work $= \frac{1}{6}$, Sita's $= \frac{1}{3}$; Together in one day $= \frac{1}{2}$ (1); Days $= 1 \div \frac{1}{2}$ (1); $= 2$ days (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	(i) Forty-five lakh sixty-seven thousand eight hundred ninety (1). (ii) 4,567,890 (1). (iii) 5,00,000 (five lakh) (1). (iv) Nearest lakh: 46,00,000 (1); new population $= 45{,}67{,}890 + 3{,}33{,}110 = 49{,}01{,}000$ (1).	1+1+1+2
34	(i) $3 \times 25 + 2 \times 10$ (1). (ii) $75 + 20 = 95$ (1). (iii) $100 - 95 = ₹5$ (1). (iv) Cost $= 5 \times 25 + 2 \times 10 = 125 + 20 = 145$ (1); change $= 200 - 145 = ₹55$ (1).	1+1+1+2
35	Net gain per minute $= 7.5 - 2.5 = 5$ litres (2); Time $= 60 \div 5 = 12$ minutes (3).	2+3
36	(i) $4(15) + 3 = 63$ (1); (ii) Figure 1 $= 7$ and figure 2 $= 11$ (1); Total $= 18$ (1); (iii) $63 - 59 = 4$ more tiles (2).	1+1+1+2
	OR: Total parts $= 6$ and $180^\circ \div 6 = 30^\circ$ (1); The angles are 30°, 60°, 90° (2); Kinds: 30° acute, 60° acute, 90° right (2).	1+2+2
37	$69 + 96 = 165$; $165 + 561 = 726$; $726 + 627 = 1353$; $1353 + 3531 = 4884$ (1); Steps $= 4$ (2); Palindrome $= 4884$ (2).	1+2+2
	OR: Exterior angles add up to 360°; one part $= 360^\circ \div 12 = 30^\circ$ (1); Exterior angles 90°, 120°, 150° (1); Interior angles 90°, 60°, 30° (1); Check: 180° (1); The triangle is right-angled (1).	1+1+1+1+1
38	Sold $= 60$ acres; remainder $= 180$ (1); Given to son $= 60$ acres; remainder $= 120$ (1); Sugarcane $= \frac{3}{4} \times 120 = 90$ acres (1); Unplanted $= 120 - 90 = 30$ acres (1); As a fraction of the whole land, unplanted $= \frac{1}{8}$ (1).	1+1+1+1+1
	OR: Larger $= (1{,}25{,}00{,}000 + 15{,}00{,}000) \div 2 = 70{,}00{,}000$ (1); Smaller $= 55{,}00{,}000$ (1); Words: seventy lakh; fifty-five lakh (1); Check: sum and difference match (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–8) | Class 7 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** 110°	Complement $= 90^\circ - a$; supplement of that $= 180^\circ - (90^\circ - a)$.
2	**(c)** 5	Repeat the routine and count.
3	**(a)** 90°	Reason with the angle sum and exterior angle property.
4	**(b)** 30	Time for the whole = time for the part ÷ fraction.
5	**(d)** 48	Divide.
6	**(b)** 2277	Distributive property.
7	**(b)** 1.7	Work out the bracket first.
8	**(b)** 55	Substitute, multiply, then halve.
9	**(d)** 290°	A complete turn is 360°.
10	**(c)** 9	Follow the rule term by term.
11	**(d)** 80°	Find x from the angle sum, then compare the angles.
12	**(b)** $\frac{1}{3}$	Unpainted = (unpainted before) × (what is not painted of it).
13	**(a)** 43,211	Round up to the next lakh.
14	**(d)** 100	Distributive property.
15	**(c)** 0.95	Find how far each number is from 1.
16	**(c)** $2n + 4$	Test the expressions with small values of n.
17	**(a)** Both A and R are true and R is the correct explanation of A.	90 - 35 = 55.
18	**(a)** Both A and R are true and R is the correct explanation of A.	2m + 1 + 2n + 1 = 2(m + n + 1), which is even.
19	**(a)** Both A and R are true and R is the correct explanation of A.	3 + 4 = 7 < 8, so the rule fails.
20	**(a)** Both A and R are true and R is the correct explanation of A.	6/12 = 1/2.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Check where the equal parts lie in each triangle; No: SSA: the equal angle is not between the equal sides, so this is not a congruence rule (1); A different triangle with the same given parts can be drawn (1).	1+1
22	Sum $= 1\frac{1}{6}$; Difference $= \frac{1}{2}$ (1); Quotient $= 2\frac{1}{3}$; Final $= 1\frac{1}{6}$ (1).	1+1
23	Compare digit counts and then place values (1); Greater: 98,76,543; difference $9,000$ (1).	1+1
24	$= 37 \times (48 + 52)$ (1); $= 37 \times 100 = 3700$ (1).	1+1
25	Largest: 85.42 (digits in decreasing order); Smallest: 24.58 (digits in increasing order) (1); Difference $= 85.42 - 24.58 = 60.84$ (1).	1+1
26	$2n + 2(n + 1) + 2(n + 2)$ (1); $= 6n + 6 = 6(n + 1)$ (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Compare numbers digit by digit (1); Descending: 12,43,567, 12,34,765, 12,34,567 (1); Difference $= 9,000$ (1).	1+1+1
28	First $= 180$ (1); Second $= (12 + 2) \times (15 - 2) = 182$ (1); Difference $= 2$ (1).	1+1+1
29	They separate by $3.5 + 4.5 = 8$ km every hour (1); After 2.5 hours: $8 \times 2.5 = 20$ km (1); Time $= 24 \div 8 = 3$ hours (1).	1+1+1
30	Numbers: $n, n + 1, \ldots, n + 2$ (1); Sum $= 3n + 3$ (1); At $n = 7$: 24 (1).	1+1+1
31	One part $= 180^\circ \div 5 = 36^\circ$ (1); Smaller $= 36^\circ$ and larger $= 144^\circ$ (1); Complement of the smaller $= 54^\circ$ and supplement of the larger $= 36^\circ$ (1).	1+1+1

	OR: $5321 - 1235 = 4086$; $8640 - 0468 = 8172$; $8721 - 1278 = 7443$; $7443 - 3447 = 3996$; $9963 - 3699 = 6264$; $6642 - 2466 = 4176$; $7641 - 1467 = 6174$ (1); Steps $= 7$ and the sequence is $1235, 4086, 8172, 7443, 3996, 6264, 4176, 6174$ (2).	1+2
32	Base b, equal sides $b + 2$; $b + 2(b + 2) = 40$, so $3b = 36$ and $b = 12$ (1); Sides $14, 14, 12$ cm (1); $12 + 14 = 26 > 14$, so the triangle exists (1).	1+1+1
	OR: First distance $= 9$ km; Second distance $= 15$ km (1); Total $= 24$ km in 6 hours (1); Average speed $= 4$ km/h (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	(i) $85.5 \times 2.5 = 213.75$, i.e. ₹213.75 (1). (ii) $250 - 213.75 =$ ₹36.25 (1). (iii) $85.5 \times 4 =$ ₹342 (1). (iv) $513 \div 85.5 = 6$ m (2).	1+1+1+2
34	(i) $2 \times 5 + 1 = 11$ sticks (1). (ii) $2n + 1$ (1). (iii) $2 \times 10 + 1 = 21$ sticks (1). (iv) $2n + 1 = 41$ gives $2n = 40$, so $n = 20$ triangles (2).	1+1+1+2
35	Brother's money $= 2x + 15$ (1); Total $= x + (2x + 15) = 3x + 15$ (1); At $x = 40$: brother $=$ ₹95 (1); Total $=$ ₹135 (2).	1+1+1+2
36	They are supplementary: one part $= 180^\circ \div 5 = 36^\circ$ (1); The angles are 72° and 108° (1); Four of the eight angles equal the smaller angle (vertically opposite, corresponding and alternate angles of it) (1); The other four equal 108° (linear pairs) (2).	1+1+1+2
	OR: $10a + b = 2(a + b)$ (1); Simplify: $8a = b$ (1); Check digit pairs: numbers 18 (1); Number of such numbers $= 1$; their sum $= 18$ (2).	1+1+1+2
37	Corresponding parts of congruent triangles are equal (1); $3x + 10 = 5x - 20$, so $2x = 30$ and $x = 15$ (1); $\angle A = 55^\circ$ (1); $2y + 4 = 3y - 6$, so $y = 10$ and $AB = 24$ cm (1); $\angle C = 180^\circ - 55^\circ - 40^\circ = 85^\circ = \angle F$ (1).	1+1+1+1+1
	OR: A: $\frac{1}{6}$, B: $\frac{1}{12}$ per hour (1); Drain: $\frac{1}{24}$ per hour (emptying) (1); Net per hour $= \frac{5}{24}$ (1); Time $= 1 \div \frac{5}{24} = 4\frac{4}{5}$ hours (2).	1+1+1+2
38	Monthly $= 4,20,000 \times 150 = 6,30,00,000$ (1); Yearly $= 75,60,00,000$ (1); Words: seventy-five crore sixty lakh (1); Nearest crore: $76,00,00,000$ (1); Show working with place values (1).	1+1+1+1+1
	OR: Income $= 50 \times 4 \times 12 + 30 \times 3 \times 20$ (1); $= 4200$ (1); Expenses $= 700 + 500 = 1200$ (1); Profit $= 3000$ (1); Expression uses brackets correctly (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–8) | Class 7 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** 125	Divide by a lakh.
2	**(d)** 130	Add the decreasing distances.
3	**(d)** 2.65	Subtract the smaller number from the larger.
4	**(c)** 22	Substitute carefully; xy means $x \times y$.
5	**(b)** 80°	Larger + smaller = 180°.
6	**(a)** 21	Triangular numbers: $n(n+1)/2$.
7	**(a)** 10	Base = $P - 2s$.
8	**(d)** 2	Multiply the fraction by the mixed number.
9	**(b)** 10	Multiply and count.
10	**(b)** 3	Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction.
11	**(b)** $\frac{1}{8}$	Write over 1000, 100 or 10 and reduce.
12	**(b)** 23	Apply the rule to the input.
13	**(b)** 120°	Total = 180° in 3 equal parts.
14	**(d)** 3	Add the reverse each time.
15	**(d)** 30°	They add up to 90°.
16	**(d)** 1	Find x first.
17	**(b)** Both A and R are true but R is not the correct explanation of A.	1,000,000 = 10,00,000 = 10 lakh, and 1 billion = 1000 million. Both true, but R does not explain A.
18	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but R does not explain A.
19	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; a zero at the end of a decimal part does not change the value, but R does not explain it.
20	**(b)** Both A and R are true but R is not the correct explanation of A.	2 x 4 + 3 = 11. Both are true, but R does not explain the value.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Let the complement be y; the angle is $3y + 6$ and $y + (3y + 6) = 90$; $4y = 84$, so $y = 21$ (1); The angle is 69° (1).	1+1
22	If n is even, $n(n+1)$ is even × odd, so even; If n is odd, $n+1$ is even, so the product is even (1); For $n = 13$: $13 \times 14 = 182$, which is even (1).	1+1
23	$\angle ACB = 180^\circ - 115^\circ = 65^\circ$; $\angle B = \angle ACB = 65^\circ$ (equal sides) (1); $\angle A = 180^\circ - 2 \times 65^\circ = 50^\circ$ (1).	1+1
24	Left = $1 - \frac{2}{7} = \frac{5}{7}$ of the number; Number = $35 \div \frac{5}{7} = 49$ (1); Removed = 14 (1).	1+1
25	1 lakh more = $1, 24, 45, 678$ (1); 1 crore less: $24, 45, 678$ (1).	1+1
26	First = 150; Second = 154 (1); First is smaller (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Distance = $300000 \times 8 = 24,00,000$ km (1); In words: twenty-four lakh (2).	1+2
28	Outer area = $(20 + 2 \times 2) \times (12 + 2 \times 2) = 384$ (1); Garden area = $20 \times 12 = 240$ (1); Path area = 144 m^2 (1).	1+1+1
29	Price per kg = $135 \div 2.5 = 54$ (1); Cost of 4.5 kg = $54 \times 4.5 = 243$ (2).	1+2
30	A: $20 + 15k$ and B: $60 + 10k$ (1); For 6 km: A = 110 and B = 120 (1); Taxi A is cheaper by ₹10 (1).	1+1+1
31	Other angle = $180^\circ - 40^\circ = 140^\circ$ (1); Halves: 20° and 70° (1); Angle between the bisectors = $20^\circ + 70^\circ = 90^\circ$ (1).	1+1+1
	OR: Try 2, 3, 4, ... numbers in turn (1); $7 + 8$; $4 + 5 + 6$; $1 + 2 + 3 + 4 + 5$ (1); Number of ways = 3 (1).	1+1+1

32	Third side between 13 and 29; Multiples of 5 in this range: $15, 20, 25$ (1); Largest third side 25 cm (1); Largest perimeter $= 8 + 21 + 25 = 54$ cm (1).	1+1+1
	OR: Pieces $= 20 \div \frac{5}{3} = 12$; Full pieces $= 12$ (1); Used $= 20$ m; Left $= 0$ m (1); Final statement (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	(i) $65°$ (1). (ii) $65°$ (1). (iii) $180° - 65° = 115°$ (1). (iv) Four angles are $65°$ (1) and four angles are $115°$ (1).	1+1+1+2
34	(i) $4 + 5 + 6$ (1). (ii) $6 + 7 + 8 = 21$ (1). (iii) Yes; the sum is $(n-1) + n + (n+1) = 3n$ (1). (iv) $10 + 11 + 12 = 33$ (1) and $16 + 17 = 33$ (1).	1+1+1+2
35	Total parts $= 9$ and $180° \div 9 = 20°$ (1); The angles are $40°$, $60°$, $80°$ (2); Kinds: $40°$ acute, $60°$ acute, $80°$ acute (2).	1+2+2
36	$155 + 551 = 706$; $706 + 607 = 1313$; $1313 + 3131 = 4444$ (1); Steps $= 3$ (2); Palindrome $= 4444$ (2).	1+2+2
	OR: Exterior angles add up to $360°$; one part $= 360° \div 15 = 24°$ (1); Exterior angles $96°$, $120°$, $144°$ (1); Interior angles $84°$, $60°$, $36°$ (1); Check: $180°$ (1); The triangle is acute-angled (1).	1+1+1+1+1
37	Sold $= 144$ acres; remainder $= 216$ (1); Given to son $= 36$ acres; remainder $= 180$ (1); Sugarcane $= \frac{4}{5} \times 180 = 144$ acres (1); Unplanted $= 180 - 144 = 36$ acres (1); As a fraction of the whole land, unplanted $= \frac{1}{10}$ (1).	1+1+1+1+1
	OR: Larger $= (98,00,000 + 24,00,000) \div 2 = 61,00,000$ (1); Smaller $= 37,00,000$ (1); Words: sixty-one lakh; thirty-seven lakh (1); Check: sum and difference match (2).	1+1+1+2
38	First $= 120$ (1); Second $= 40$ (1); Larger: first (1); Difference 80 (2).	1+1+1+2
	OR: Total cost $= 480 + 540 = 1020$ (1); Total weight $= 5$ kg (2); Cost per kg $= 1020 \div 5 = 204$ (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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